

Research on the application of artificial intelligence in medical imaging diagnosis

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Abstract—This paper first expounds the research status for artificial intelligence technology in medical imaging diagnosis, and illustrates the importance of computer-aided diagnosis with examples; Secondly, the current bottlenecks in the development of computer-aided diagnosis technology are analyzed in detail from the aspects of technology, industry and application; Finally, based on the previous analysis, the paper puts forward some suggestions on how to better use artificial intelligence technology in medical imaging diagnosis with reference to the current actual situations.

Keywords—Artificial intelligence; Medical imaging; Computer-aided diagnosis

I. INTRODUCTION

According to the statistics of the World Health Organization (WHO) [1], among the deaths caused by cancer, cardiovascular disease, diabetes, and chronic respiratory diseases in the world, cancer mortality ranks first, and the number of deaths from the disease accounts for 22.32% of the total number of deaths. In recent years, the global incidence of cancer has shown a high trend, and the number of diagnosed patients has increased year by year, from 17.2 million in 2016 to 19.3 million in 2020, and is expected to increase to 20.2 million in 2022. More seriously, the incidence of cancer in my country has become increasingly younger [2]. One of the main reasons is the unbalanced economic and social development in various regions, especially in some rural areas with backward medical technology and high treatment costs. Many tumors have missed the best time for treatment after being investigated and dealt with, resulting in a long-term high mortality rate. Therefore, it is necessary to conduct regular inspections of related diseases, so that the diseased part can be detected in

time and the corresponding treatment can be carried out to increase the chance of survival.

Nowadays, while information technology and technological innovation gradually change people's way of life, the integration of computer technology and other fields also promotes the development of society. In this context, medical imaging technology [3] has been widely used in the screening of a variety of diseases. It mainly refers to the technical means and process of directly obtaining internal tissue images by non-penetrating methods for a certain part of the body and the body; According to statistics [4], medical imaging technology has been widely used in more than 70% of clinical diagnosis. It mainly has the following three functions in clinical practice: auxiliary medical clinical examination (such as identification, marking, determination and classification), injection processing method determination (such as cutting, stroking, sizing and evaluation) and interventional therapy guidance (e.g., 3D visualization). At present, medical imaging detection methods have been diversified [5], such as CT (computerized cross-sectional image), CR (computer X-ray photography), MRI (magnetic resonance image), PET-CT (positron emission tomography), DSA (angiography), ultrasound, and endoscopy, etc. Generally, the pathological process of most diseases easily leads to changes in human physiology, and such changes will also form different imaging information in different imaging examinations. Effective analysis of these information allows physicians to identify potential causes and monitor the status of related diseases, which in turn can determine subsequent treatment and diagnostic measures.

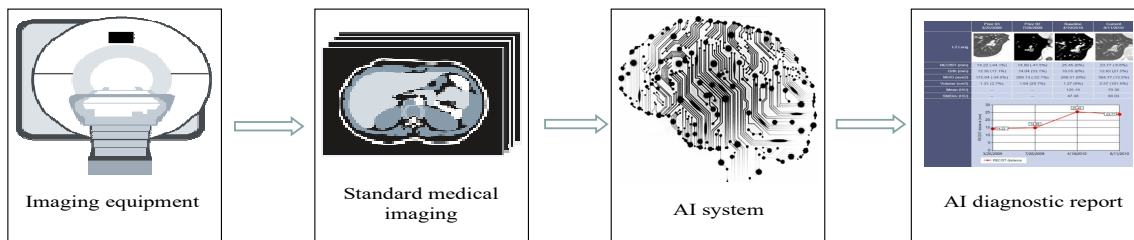


Figure 1. The overall process of AI-based computer-aided diagnosis.

However, with the increasing number of hospital patients, the contradiction between supply and demand of medical image analysis continues to increase. In addition, due to the lack of diagnosis and treatment experience of primary-level physicians, the efficiency of manual reading is prone to be low. In view of this, the emergence of computer-aided diagnosis (CAD) [6] can help doctors to complete the diagnosis and treatment of diseases, and help doctors to obtain image information more quickly and make certain qualitative and quantitative analysis. It has shown great positive effects in intelligent medicine, improving the efficiency and accuracy of doctors' diagnosis. Fig. 1 shows the overall process of AI-based CAD. Many researchers at home and abroad attach great importance to the design process of CAD system and devote themselves to the application of this technology in clinical diagnosis to a great extent.

The main task of CAD system is to help doctors to

make a clear division and accurate identification of the lesion area. The time-saving and efficient detection helps patients to discover their own diseases in time, proving that the system is suitable for clinical diagnosis of diseases. Therefore, the combination of computer technology and diagnostic analysis of medical image has an important role in improving the efficiency and accuracy of lesion detection. At present, traditional medical image analysis algorithms have been widely used in CAD system, but these methods have some difficult problems in both process and technology. For example, traditional segmentation algorithms are designed based on low-level information and fail to make full use of high-level semantic information, thus failing to extract rich edge and texture features of images, resulting in poor diagnostic results. It can be seen from Fig. 2 that AI-based methods are faster and more efficient than traditional methods for disease diagnosis.

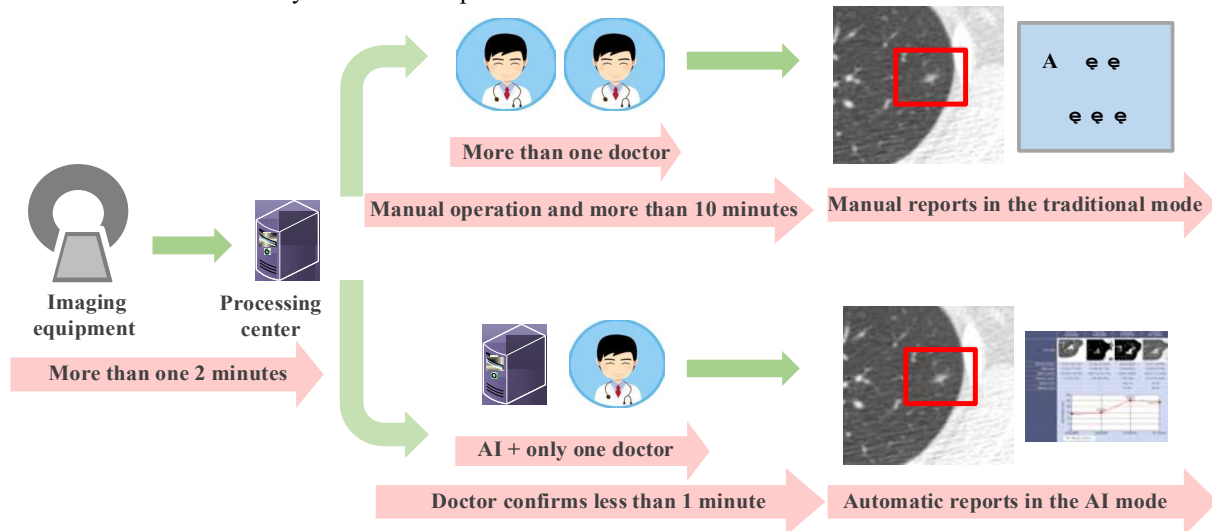


Figure 2. Comparison between traditional and AI-based methods for disease diagnosis.

With the rapid development of software and hardware technology, the artificial intelligence (AI) medical imaging technology based on deep learning [7] overcomes the shortcomings mentioned above. In recent years, deep learning technology has achieved great success in the field of computer vision, with overwhelming advantages in classification, segmentation and object detection. In particular, convolutional neural networks (CNNs) can accurately identify the location of lesions and obtain the information of lesions by combining deep learning technology with medical imaging knowledge, providing quantitative diagnostic basis for subsequent treatment plans of patients. It can automatically extract visual features through continuous image training so as to realize the recognition and disease diagnosis of diversified medical images.

At present, AI technology can complete tasks such as automatic identification and labeling of lesions, automatic

delineation of target areas and 3D reconstruction of medical images through image classification, image segmentation, target detection and image retrieval. It is mainly applied in the stages of screening, diagnosis and treatment of diseases. Therefore, AI brings many benefits to medical image analysis: firstly, it can process and analyze images quickly, and give auxiliary judgment results in time; Secondly, it has good diagnostic sensitivity, which can effectively reduce the missed diagnosis rate; Thirdly, it can carry out accurate data analysis to bridge the gap in skills and experience among physicians, thereby improving the quality of primary disease screening. Fig. 3 shows deep learning-based CAD results for various medical images, in which the color-labeled parts represent the identified lesion regions. Specifically, Fig. 3 (a)-(f) denote AI-based lung nodule examination, liver tumor screening, MRI stroke screening, MRI prostate screening, chest X-ray examination and mammography screening, respectively.

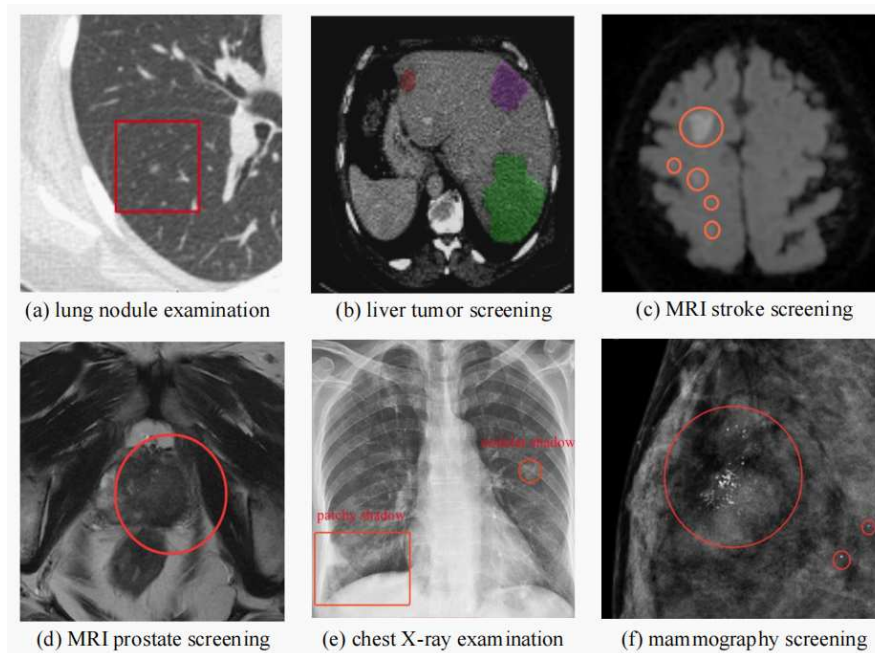


Figure 3. Results of CAD using AI technology.

II. CHALLENGES FACED BY AI MEDICAL IMAGING

With the continuous optimization of software and hardware, deep learning technology has achieved great success in the diagnosis of medical imaging. Furthermore, the number of companies and institutions joining this field is increasing, and market investment is also constantly upgrading. However, there is still a certain gap between AI-based medical imaging diagnosis technology and clinical application [8], mainly in the following three aspects:

A. It is technically difficult to obtain annotated datasets for model training.

Studies have shown that the predictive performance of deep learning-based medical image diagnostic models mainly depends on the scale of the training dataset. In general, the more samples the model is trained on, the more reliable the model predicts. However, it is still difficult to obtain high-quality datasets due to the following reasons: First, the standardization of medical imaging data is low and digital imaging data is difficult to obtain; Secondly, the labeling of data sets is difficult. Because of the professionalism of medicine itself, the threshold for people who can participate in labeling is very high, and the quality of labeling cannot be guaranteed; Furthermore, the definition of many problems in clinical medicine is still ambiguous.

B. There is a lack of certain evaluation standards for AI-assisted diagnosis results in the industry.

The evaluation indicators for screening or diagnosis of different diseases should belong to their own unified standards, and the evaluation indicators for different diseases should be different to avoid confusion or mixed use of indicators. Secondly, the legal and regulatory system

related to registration, access and supervision has not yet been formed. Although the popularity of the Chinese market is no less than that of Western countries, the follow-up of laws and regulations is relatively slow. For example, the access standards and evaluation methods for artificial intelligence software products are still in the blank stage, which has also become an important bottleneck restricting related companies from realizing product optimization.

C. There are still some limitations for AI to solve medical imaging problems in practical applications.

Since deep learning has its own characteristics of self-learning and self-adaptation [9], it requires the excessive learning from a large amount of sample data to obtain richer image features for further disease diagnosis; Secondly, due to the complex process of medical image information processing, it is difficult to build a reliable model and train images with small differences in information and diverse structures and shapes (such as organs and dynamic images), resulting in unsatisfactory results.

III. SEVERAL SUGGESTIONS FOR AI MEDICAL IMAGING

Referring to the comprehensive analysis discussed above, how to quickly apply the deep learning-based computer-aided diagnosis technology to clinical diagnosis [10] will be an urgent need for smart medical care to take root. Based on this, the paper puts forward some suggestions on the application of artificial intelligence in medical imaging diagnosis according to the actual situation:

A. Technical Advices

It is particularly important to technically optimize the deep learning algorithm model and establish a high-quality data set [11], which mainly includes: one is to deepen the research on the joint model of rules and deep learning; The other is to establish a wide range of sources and standardized data sets to improve the quality of training data; More importantly, transfer learning can be used to reduce annotations in model training to improve annotation efficiency.

B. Advices on the AI industry

More efforts should be made to investigate and establish registration specifications for AI software in the industrial field. The main content is to focus on the model's pre-, mid- and late-stage process specifications, model performance evaluation indicators, model safety evaluation, diagnostic recommendation grading system and medical institution access mechanism; And then, relevant institutions should establish a new professional team of artificial intelligence medical application research, whose main task is to study the model evaluation framework and indicators; Finally, professional organizations such as expert alliances related to imaging medicine should study and formulate AI-based medical imaging and related professional consensus, and provide professional medical guidance and advice for enterprises or customers.

C. Advices on the AI application

According to the application specifications of medical images in the clinical diagnosis and treatment process, the standardization and standardized training of digital annotation should be strengthened to improve the interoperability of data output in the CAD system. The training of medical imaging diagnosis models based on deep learning should not only focus on the identification and classification of image signals, but should also make comprehensive judgments based on the patient's medical history, symptoms, and other test results. In this way, the overall medical level of artificial intelligence can be further improved, and then the transition from auxiliary examination to predictive diagnosis and auxiliary judgment can be realized.

IV. CONCLUSION

The paper comprehensively analyzes the difficulties existing in the current deep learning-based artificial intelligence technology in medical image analysis, and puts forward corresponding suggestions from the aspects of technology, industry and application. In general, although the AI-based medical imaging computer-aided diagnosis mode can effectively reduce the work intensity of front-line clinical imaging doctors, a lot of practice and exploration are still needed to further improve the effectiveness and reliability of the entire disease diagnosis process. Therefore, how to improve the accuracy of lesion area detection, ensure the

detection quality of the auxiliary diagnosis system and provide correct disease diagnosis information, in order to further improve the effectiveness and reliability of the whole diagnosis process, will become the main direction of future research.

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REFERENCES

- [1] Sung H, Ferlay J, Siegel R L, et al. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries[J]. CA: a cancer journal for clinicians, 2021, 71(3): 209-249.
- [2] Clinton S K, Giovannucci E L, Hursting S D. The world cancer research fund/American institute for cancer research third expert report on diet, nutrition, physical activity, and cancer: impact and future directions[J]. The Journal of nutrition, 2020, 150(4): 663-671.
- [3] Chandy A. A review on iot based medical imaging technology for healthcare applications[J]. Journal of Innovative Image Processing (JIIP), 2019, 1(01): 51-60.
- [4] Smith N B, Webb A. Introduction to medical imaging: physics, engineering and clinical applications[M]. Cambridge university press, 2010.
- [5] Kasban H, El-Bendary M A M, Salama D H. A comparative study of medical imaging techniques[J]. International Journal of Information Science and Intelligent System, 2015, 4(2): 37-58.
- [6] Doi K. Computer-aided diagnosis in medical imaging: historical review, current status and future potential[J]. Computerized medical imaging and graphics, 2007, 31(4-5): 198-211.
- [7] Zhou S K, Greenspan H, Davatzikos C, et al. A review of deep learning in medical imaging: Imaging traits, technology trends, case studies with progress highlights, and future promises[J]. Proceedings of the IEEE, 2021, 109(5): 820-838.
- [8] Han C, Rundo L, Murao K, et al. Bridging the gap between AI and healthcare sides: towards developing clinically relevant AI-powered diagnosis systems[C]//IFIP International Conference on Artificial Intelligence Applications and Innovations. Springer, Cham, 2020: 320-333.
- [9] Janiesch C, Zschech P, Heinrich K. Machine learning and deep learning[J]. Electronic Markets, 2021, 31(3): 685-695.

- [10] Fujita H. AI-based computer-aided diagnosis (AI-CAD): the latest review to read first[J]. Radiological physics and technology, 2020, 13(1): 6-19.
- [11] Cho J, Lee K, Shin E, et al. How much data is needed to train a medical image deep learning system to achieve necessary high accuracy?[J]. arXiv preprint arXiv:1511.06348, 2015.